DAL-DETR: Dynamic Adaptive Lightweight Detection Transformer for Small Object Detection in Low-Altitude Remote Sensing Imagery

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***Abstract*—Small object detection in low-altitude remote sensing imagery plays a pivotal role in urban planning, disaster assessment, and related applications. However, existing approaches commonly suffer from excessive model complexity and substantial computational overhead, making it challenging to achieve an optimal balance between lightweight design and detection accuracy on resource-constrained edge computing devices. To address these limitations, we propose dynamic adaptive lightweight DETR (DAL-DETR). To overcome the constraints of fixed receptive fields in conventional convolutions—which struggle to simultaneously capture local details and global contextual semantics of small objects—we design a dynamic adaptive mixer backbone (DAMB). This module leverages dynamic multibranch convolutions (DMCs) to achieve adaptive fusion of multiscale receptive fields, improving multiscale feature representation while significantly reducing parameter count. To mitigate computational redundancy introduced by feature concatenation, we introduce a lightweight dynamic fusion neck (LDFN), which replaces traditional concatenation operations with deformable modulation fusion, further reducing model complexity and computational cost. In addition, we incorporate an auxiliary decoupled geometric loss (ADGL) that employs an auxiliary bounding box generation mechanism with decoupled geometric constraints, improving localization precision for small objects. Extensive experiments on multiple public datasets demonstrate that DAL-DETR achieves a 45.2% reduction in parameters and a 34.0% decrease in computational cost compared to the baseline, while still delivering consistent accuracy improvements and outperforming existing state-of-the-art detection methods.**

***Index Terms*—Feature pyramid network (FPN), lightweight network, low-altitude remote sensing, small object detection, Transformer.**

I. Introduction

With the rapid advancement of uncrewed aerial vehicle (UAV) and satellite remote sensing technologies, low-altitude remote sensing has emerged as an increasingly vital tool across diverse domains including urban planning, traffic monitoring, disaster assessment, and military reconnaissance. As a fundamental task in low-altitude remote sensing image analysis, object detection performance directly determines the effectiveness and reliability of downstream applications. However, object detection in low-altitude remote sensing imagery confronts distinctive challenges: due to elevated capture altitudes, targets typically occupy limited pixels and are highly susceptible to complex background noise interference [1], exhibiting characteristics of diminutive size and drastic scale variations. More critically, edge computing platforms such as UAVs impose stringent computational resource constraints on detection models, necessitating high computational efficiency while maintaining accuracy [2]. Consequently, achieving high-precision small object detection in low-altitude remote sensing under limited computational budgets has become a pressing challenge in this field.

Convolutional neural network (CNN)-based object detection methods have achieved remarkable progress over the past decade [3]. Two-stage detectors such as Faster R-CNN [4] and Cascade R-CNN [5] are renowned for their superior detection accuracy, yet their intricate detection pipelines constrain their deployment in resource-constrained scenarios. Single-stage detectors represented by the YOLO series [6], [7], [8], [9] have substantially improved inference speed through end-to-end architectures, but remain reliant on nonmaximum suppression (NMS) and manually designed label assignment strategies [10]. Moreover, the localized receptive fields inherent to conventional convolutions limit their capacity to capture global contextual information [11], rendering them ineffective in modeling complex geometric features such as elongated objects in low-altitude remote sensing imagery [12].

Transformer-based end-to-end detectors have introduced novel solutions to these challenges. Detection Transformer (DETR) [13] eliminated postprocessing steps like NMS through bipartite graph matching, albeit suffering from slow training convergence. Subsequently, methods including Deformable-DETR [14], DN-DETR [10], and DINO [15] have progressively refined query mechanisms and enhanced detection accuracy, though at the cost of increased model complexity. The introduction of RT-DETR [16] marked the first realization of real-time end-to-end object detection within the DETR family. In low-altitude remote sensing scenarios, approaches such as Enhanced Swin Transformer [17] and TRD [18] have improved feature perception capabilities, yet typically incur substantial model complexity [2], presenting

Received 13 April 2026; revised 2 May 2026 and 31 May 2026; accepted 20 June 2026. Date of publication 24 June 2026; date of current version 1 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 52575590 and Grant 51905378, in part by the Seed Foundation of Tianjin University under Grant 1354, in part by Tianjin Science and Technology Plan Project under Grant 22JCYBJC01670 and Grant 21JCZDJC00820, and in part by the Natural Science Foundation of Tianjin under Grant 20JCQNJC00360. *(Corresponding author: Yang Qi.)*

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Digital Object Identifier 10.1109/TGRS.2026.3706691

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ongoing challenges for deployment in resource-constrained environments.

Small object detection constitutes one of the central challenges in low-altitude remote sensing image analysis. Given that small objects occupy minimal pixel coverage, their feature representations are inherently weak and prone to background noise interference, rendering effective feature extraction critically important. Multireceptive field feature extraction modules [19] enhance feature representations by aggregating contextual information from multiple receptive fields, while LSKNet [20] dynamically adjusts receptive fields to accommodate background variations. Nevertheless, the fragile features of small objects are susceptible to progressive attenuation or even complete loss during network propagation, necessitating feature pyramid networks (FPNs) capable of effectively fusing multiscale information to compensate for feature degradation. FPN [21] pioneered the use of top-down pathways for semantic information propagation, while path aggregation network (PANet) [22] and bidirectional FPN (BiFPN) [23] established alternative information flow architectures. Recently, CAEM-DETR [24] and SAFF-DETR [25] have improved small object detection performance through contrastive attention mechanisms and frequency-domain feature fusion, respectively. However, these approaches typically introduce substantial additional parameters while strengthening feature representations, resulting in significantly elevated model complexity. Furthermore, even with improved feature extraction and fusion, bounding box regression for small objects remains challenging—owing to their diminutive size, bounding boxes exhibit extreme sensitivity to positional perturbations, and existing IoU-based loss functions fail to adaptively modulate optimization intensity according to regression sample quality. Although GALDET [28] and YOLO26 [9] have enhanced small object localization accuracy through refined loss functions, this issue remains inadequately addressed.

To address the limitations of existing methods in balancing model efficiency and detection accuracy, we propose dynamic adaptive lightweight DETR (DAL-DETR), an efficient network for low-altitude remote sensing object detection. This architecture achieves efficient fusion of multimorphology receptive fields through a dynamic adaptive mixer backbone (DAMB), subsequently employing a lightweight dynamic fusion neck (LDFN) that replaces feature concatenation with modulated fusion, effectively reducing computational overhead in cross-scale feature integration. For bounding box regression optimization, an auxiliary bounding box generation mechanism coupled with decoupled geometric constraints is introduced to enhance small object localization precision. These designs enable DAL-DETR to simultaneously reduce model complexity while improving detection accuracy.

The primary contributions of this work are summarized as follows.

1) To address the challenges of high model complexity and low computational efficiency in low-altitude remote sensing object detection, we propose DAL-DETR, an efficient network tailored for low-altitude remote sensing scenarios. This approach achieves multimorphology receptive field fusion through dynamic multibranch convolutions (DMCs), employs modulation strategies to replace feature concatenation for reduced fusion overhead, and incorporates an auxiliary bounding box mechanism to optimize the regression process, thereby achieving improved detection accuracy with substantially reduced parameters and computational cost.

2) To resolve the limitations of fixed-size convolutional kernels that provide singular receptive fields and struggle to simultaneously capture local details and contextual semantics of small objects, we propose DAMB. Motivated by the directional structural features prevalent in low-altitude remote sensing scenes, such as roads, vehicle arrays, and bridges, DAMB introduces orthogonal strip convolution kernels alongside square kernels, combined with an adaptive scale mixer (ASM) to construct hierarchical multiscale representations, enhancing multiscale feature expressiveness while reducing parameter count.

3) To overcome the channel dimension doubling and excessive computational overhead caused by feature concatenation fusion strategies, we propose LDFN. By replacing channel concatenation with modulated fusion, this neck adaptively filters low-level features using high-level semantic information and introduces a dynamic point sampling mechanism for efficient learnable upsampling, effectively compressing model scale.

4) To strengthen bounding box regression accuracy for small objects, we propose an auxiliary bounding box generation mechanism that refines the network's optimization process. In addition, decoupled geometric constraint terms are introduced to effectively optimize bounding box predictions from a shape feature perspective, improving convergence quality in small object bounding box regression.

The source code is available at https://github.com/Huang27-2/DAL-DETR

II. Related Work

A. CNN-Based Object Detection

Recent years have witnessed remarkable advances in CNN-based object detection, which can be categorized into two-stage and one-stage paradigms depending on their detection pipelines [3]. Two-stage detectors are renowned for their superior accuracy. R-CNN [30] pioneered the integration of deep learning into object detection tasks, while Faster R-CNN [4] introduced the region proposal network (RPN) to enable end-to-end prediction of candidate regions. Cascade R-CNN [5] further enhanced detection precision through a cascaded regression framework that progressively refines predictions. However, two-stage approaches face challenges in computational efficiency and model complexity, making them less suitable for deployment on resource-constrained platforms such as UAVs.

To address real-time requirements, one-stage detectors have emerged as a compelling alternative. The YOLO series [6], [7], [8], [9] directly detects objects in an end-to-end manner, eliminating redundant computational overhead and establishing itself as the dominant solution for real-time detection. Nevertheless, these methods exhibit several inherent limitations. First, they demonstrate sensitivity to anchor generation

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strategies and rely on manually designed components such as NMS and label assignment rules [10], which not only complicate hyperparameter tuning but also introduce additional inference latency. Second, the intrinsic local receptive fields of conventional convolutions constrain their capacity to capture global contextual information [11], resulting in suboptimal localization and classification accuracy in complex scenarios. Moreover, in low-altitude remote sensing imagery, objects undergo dramatic scale variations and frequently exhibit elongated geometries, which standard convolutions struggle to model effectively [12].

B. Transformer-Based Object Detection

To overcome the dependence on handcrafted components and limited global modeling capabilities of CNN-based detectors, Transformer-based end-to-end object detectors have opened new research directions. DETR, proposed by Carion et al. [13], represents the first Transformer-based end-to-end detector, which directly predicts one-to-one object sets through bipartite matching, eliminating postprocessing steps like NMS and streamlining the detection pipeline. However, the original DETR confronts two major challenges: slow training convergence and difficulties in query optimization.

Subsequently, numerous improvements have been proposed to address these issues. Deformable-DETR [14] significantly accelerates convergence by incorporating multiscale features and designing efficient deformable attention modules. Conditional DETR [31] and Anchor DETR [32] simplify query optimization through conditional spatial queries and anchor mechanisms, respectively. DAB-DETR [33] employs 4-D reference point coordinates as queries, updating bounding boxes iteratively in a cascaded manner. DN-DETR [10] introduces a query denoising training strategy that bypasses bipartite matching to directly learn approximations of ground-truth boxes, substantially accelerating training convergence. DINO [15] further advances contrastive denoising strategies, achieving state-of-the-art performance at the time.

While these methods continuously improve detection accuracy, their increased parameter counts and computational complexity pose challenges for real-time applications. RT-DETR [16] marks a milestone as the first real-time end-to-end detector in the DETR family, achieving significant inference speedup while maintaining high accuracy through redesigned encoder architecture and optimized decoder query mechanisms. With the gradual adoption of Transformers in low-altitude remote sensing object detection, Enhanced Swin Transformer [17] improves feature perception through a locality-aware backbone, TRD [18] enhances detection performance using an attention-augmented backbone, and SFRNet [34] extracts fine-grained features via discrete feature refinement and multi-ROI loss functions. However, these methods often introduce parameter redundancy [2] while pursuing high accuracy, presenting ongoing deployment challenges on edge computing platforms such as UAVs.

C. Small Object Detection

Object detection within low-altitude remote sensing scenarios remains one of the field's most formidable challenges. Due to elevated capture altitudes, objects occupy only a limited number of pixels and are highly susceptible to interference from complex background noise [1], posing substantial challenges for effective feature extraction. To address this, multireceptive field feature extraction modules [19] enhance feature representation by aggregating contextual information across multiple receptive fields, while LSKNet [20] dynamically adjusts receptive fields to accommodate background variations. GMDRA-net [35] introduces a global multilevel perception module that selectively aggregates discriminative features while filtering noise. SAR-NanoShipNet [36] employs deformable convolution and corner attention mechanisms to adaptively capture multidirectional boundary features of small targets, enhancing feature expressiveness in complex backgrounds. However, these approaches often introduce considerable additional parameters when strengthening feature expressiveness, leading to substantial increases in model complexity.

Beyond this, the loss of small object features during network propagation remains an urgent problem requiring resolution. FPN [21] pioneered the use of top-down pathways to propagate semantic information, followed by PANet [22] and BiFPN [23], which constructed alternative information transmission pathways for efficient multiscale feature aggregation. CAEM-DETR [24] models positive–negative feature polarity interactions through contrastive attention mechanisms and incorporates spatial-frequency dual-domain information to enhance small object detection performance. SAFF-DETR [25] achieves efficient multiscale feature interaction via multibranch frequency feature fusion and cross-layer frequency attention mechanisms. Along similar lines, MHAF-YOLO [26] proposes a multibranch auxiliary FPN that introduces superficial assisted fusion and advanced assisted fusion modules to strengthen small object detection capability by preserving shallow spatial details and enriching multiscale feature interactions across neck layers. Furthermore, UAV-DETR [27] enhances small and occluded object detection in UAV imagery by jointly leveraging spatial and frequency domain information through a multiscale feature fusion with a frequency enhancement module while employing a semantic alignment and calibration module to address feature misalignment across fusion paths. Although these methods improve feature fusion effectiveness, their intricate network architectures incur substantial computational overhead.

Furthermore, conventional bounding box regression loss functions exhibit notable deficiencies in small object localization. Existing IoU-based loss functions primarily accelerate convergence by introducing geometric penalty terms, yet overlook the limitation that the IoU loss term itself cannot adaptively adjust optimization intensity according to regression sample quality. CGMDet [37] integrates positional information through coordinate and global information aggregation modules combined with enhanced bounding box regression loss, while GALDET [28] refines bounding box regression via an improved IoU loss function, boosting localization accuracy for small objects. YOLO26 [9] introduces progressive loss (ProgLoss) balancing and small target-aware label (STAL) assignment strategies, effectively improving

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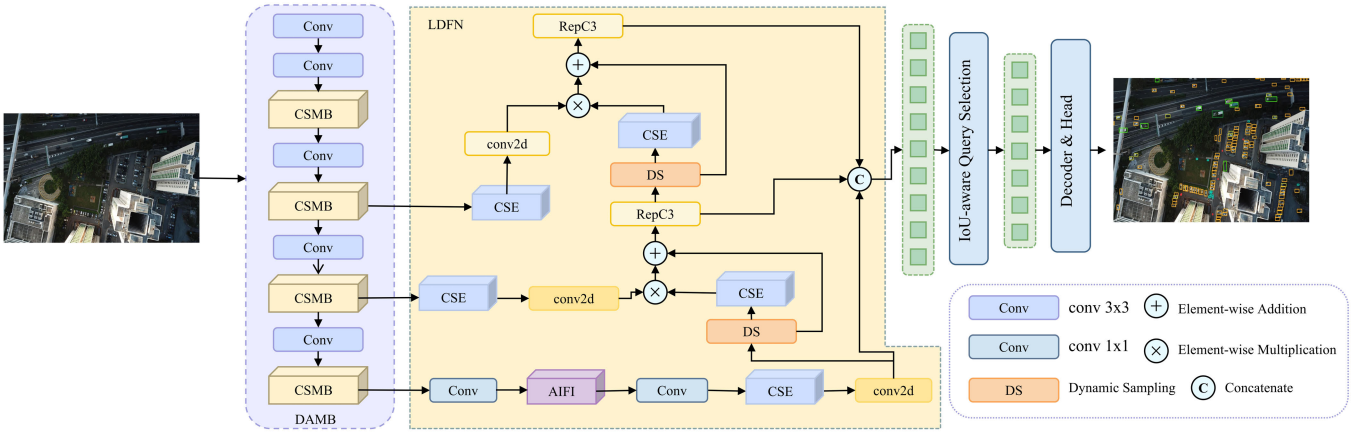


Fig. 1. Overall architecture of the proposed DAL-DETR framework.

recognition accuracy for small objects. SD loss [29] mitigates the scale sensitivity of small object bounding box regression by adaptively reweighting loss components as a function of target size, yet the intrinsic inability of the IoU loss term to elicit discriminative gradient responses across regression samples of heterogeneous quality remains unaddressed. On the contrary, the proposed auxiliary decoupled geometric loss (ADGL) modulates gradient responses according to regression sample quality via an auxiliary bounding box mechanism.

In summary, existing research on low-altitude remote sensing small object detection faces the challenge of reconciling model lightweighting with detection accuracy [2]. To overcome these limitations, this article proposes the DAL-DETR network: we design a lightweight backbone based on DMC and adaptive scale mixing mechanisms to enhance multiscale feature representation while reducing parameter count; we construct an efficient feature pyramid structure leveraging channel-adaptive weighting and learnable dynamic upsampling to achieve comprehensive fusion of shallow-level details and deep-level semantics; and we improve regression loss by incorporating auxiliary bounding boxes and decoupled geometric constraints to enhance small object localization accuracy.

III. Methodology

A. Architecture Overview

Low-altitude remote sensing small object detection confronts multiple challenges, including diminutive target dimensions, intricate backgrounds, and constrained edge computing resources. As the pioneering real-time end-to-end object detector, RT-DETR achieves an effective balance between accuracy and speed through its efficient hybrid encoder. However, limitations emerge when directly deploying it for detecting small objects in low-altitude remote sensing scenarios: the backbone network employs fixed-size convolution kernels, whose uniform receptive field configuration struggles to simultaneously capture fine-grained local details and contextual semantics of small targets; the cross-scale feature fusion module CCFF adopts a feature concatenation strategy that doubles channel dimensions, incurring substantial computational overhead; existing IoU loss functions cannot adaptively modulate optimization intensity according to regression sample quality, thereby constraining localization precision for small objects. To address these issues, we propose DAL-DETR, a lightweight network tailored for remote sensing small object detection, using RT-DETR as the baseline. The overall architecture is illustrated in Fig. 1.

The backbone network incorporates our proposed DAMB, which utilizes a cross-stage mixer block (CSMB) as the fundamental building unit. Through DMC and ASM, it achieves efficient multiscale feature extraction while reducing model complexity and preserving robust feature representation capability, ultimately feeding three-stage features into the encoder. The efficient hybrid encoder transforms multiscale features into image feature sequences via intrascale feature interaction and cross-scale feature fusion. In particular, intrascale feature interaction is accomplished through the AIFI module, while cross-scale feature fusion employs our proposed LDFN to replace the original CCFF module, further compressing model size through selective fusion with channel attention and learnable dynamic upsampling mechanisms. Subsequently, the uncertainty-minimal query selection mechanism extracts a fixed number of features from encoder outputs to serve as initial object queries for the decoder. Finally, the Transformer decoder with auxiliary prediction heads iteratively refines object queries to generate category predictions and bounding boxes. During training, we substitute the original IoU loss with ADGL, which enhances small object localization performance through an auxiliary bounding box mechanism, effectively balancing model lightweighting and accuracy without introducing additional complexity.

B. Adaptive Feature Extraction With DAMB

In low-altitude remote sensing small object detection tasks, targets exhibit minimal pixel occupancy and dramatic scale variations, imposing stringent demands on the multiscale feature extraction capability of backbone networks. However, existing methods predominantly employ fixed-size square convolution kernels, whose uniform receptive field configuration struggles to simultaneously capture fine-grained local details of small targets and broad contextual semantics. To expand

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receptive field diversity, multibranch parallel architectures have been widely adopted, yet their static branch fusion strategies cannot adaptively adjust to input content, thereby limiting feature representation flexibility.

To address these limitations, we propose DAMB. Motivated by the geometric diversity of targets in low-altitude remote sensing scenes, this architecture incorporates geometry-aware convolution kernels within a cross-stage feature reuse framework, enabling efficient multiscale feature extraction through input-aware adaptive fusion strategies.

First, we design DMC as the fundamental feature extraction component. This unit abandons the single square convolution kernel design in favor of three geometrically complementary depthwise convolutions operating in parallel. For input feature map $X \in R^{C\times H\times W}$, we configure isotropic square convolution kernels to capture local textures while introducing orthogonal strip convolution kernels to model directional structural features. This design is motivated by the observation that low-altitude remote sensing scenes contain both isotropic local textures and abundant directional structural features such as roads, vehicle arrays, and bridges. The square kernel captures rotation-invariant local patterns, while the horizontal and vertical strip kernels model directional structures along their respective orientations, achieving comprehensive multidirectional feature coverage through geometric complementarity. These three geometrically complementary depthwise convolutions together constitute the multimorphology receptive fields of DMC. The outputs of three convolution operations are formalized as

$$F_i = \mathrm{DWConv}_i(X), \quad i \in \{1,2,3\} \tag{1}$$

where $\mathrm{DWConv}_1, \mathrm{DWConv}_2, \mathrm{DWConv}_3$ correspond to $k \times k, 1 \times l,\ l \times 1$ depthwise convolution operations, respectively. Depthwise convolution maintains a linear relationship between parameters and channel count, effectively controlling computational overhead.

The fusion strategy for multibranch outputs is critical to feature representation quality. Static weight fusion overlooks the differentiated demands of various inputs for different morphological receptive fields. Therefore, we introduce a dynamic weight generation mechanism within DMC to achieve input-aware adaptive fusion. This mechanism first extracts channelwise statistical descriptors via GAP, then generates normalized fusion coefficients for each branch through $Conv_{1\times1}$

$$\mathbf{w} = \mathrm{Softmax}\left(\mathrm{Conv}_{1\times1}(GAP(\mathbf{X}))\right) \in R^{3\times C}. \tag{2}$$

Softmax normalization along the branch dimension establishes competitive relationships among the three branch weights, enabling the network to learn optimal receptive field combinations for current inputs. The fusion process is represented as

$$Y = \sigma\left(\mathrm{BN}\left(\sum_{i=1}^{3} w_i \odot F_i\right)\right) \tag{3}$$

where $\odot$ denotes elementwise multiplication, and $\sigma(\cdot)$ represents the SiLU activation function.

While DMC operates at a single scale configuration, we further design ASM to construct a hierarchical receptive field system. This module equally partitions input features along the channel dimension, processes them separately through DMC configured with different base kernel sizes, and completes cross-scale information integration via channel aggregation

$$\mathbf{Z} = \mathrm{Conv}\left(\left[\mathbf{F}_k\left(\mathbf{X}^{(1)}\right); \mathbf{F}_k\left(\mathbf{X}^{(2)}\right)\right]\right) \tag{4}$$

where $\mathbf{X}^{(1)}, \mathbf{X}^{(2)} \in R^{C/2\times H\times W}$ denotes partitioned feature subsets, $\mathbf{F}_k(\cdot)$ represents DMC with base kernel size k, and $[\cdot;\cdot]$ indicates channelwise concatenation.

Building upon this foundation, we construct the dynamic scale mixer block (DSMB) as the core processing unit of DAMB. To ensure training stability in deep networks, DSMB introduces the layer scale mechanism for channelwise scaling of residual paths and incorporates a stochastic depth strategy to enhance regularization. Its output is expressed as

$$\mathbf{X}' = \mathbf{X} + \Gamma\left(\gamma_1 \odot \mathrm{ASM}\left(\mathrm{BN}\left(\mathbf{X}\right)\right)\right) \tag{5}$$

$$\mathbf{X}^{out} = \mathbf{X}' + \Gamma\left(\gamma_2 \odot \mathrm{GLU}\left(\mathrm{BN}\left(\mathbf{X}'\right)\right)\right) \tag{6}$$

where $\gamma_1, \gamma_2 \in R^C$ represents learnable scaling factors initialized to 10^{-2}, $\Gamma(\cdot)$ denotes the stochastic depth function, and GLU represents gated linear units.

Based on the cross-stage reuse framework [38], we design CSMB as the core module of DAMB, with its structure illustrated in Fig. 2. CSMB partitions input features after channel expansion into direct paths and transformation paths, where transformation paths comprise cascaded DSMB stages, with outputs from each stage aggregated with direct features

$$\mathbf{Y}_0, \mathbf{Y}_1 = \mathrm{Split}\left(Conv_{1\times1}(\mathbf{X})\right) \tag{7}$$

$$\mathbf{Y}_{j+2} = \mathrm{DSMB}\left(\mathbf{Y}_{j+1}\right), \quad j = 0, 1, \ldots, n-1 \tag{8}$$

$$\mathbf{O} = \mathrm{Compress}\left(\left[\mathbf{Y}_0; \mathbf{Y}_1, \ldots, \mathbf{Y}_{n+1}\right]\right) \tag{9}$$

where $\mathbf{O}$ represents the final module output. This design preserves the gradient flow advantages of cross-stage architectures while fully reusing multilevel feature information through dense connections.

In summary, DAMB achieves dynamic fusion of multimorphology receptive fields through DMC, constructs hierarchical multiscale representations via ASM, and accomplishes effective propagation of multilevel semantics through cross-stage dense connections. This progressive feature extraction framework, from local to global, enables the network to effectively address scale variations and feature sparsity challenges in remote sensing small object detection while maintaining lightweight characteristics.

C. Lightweight Feature Fusion With LDFN

In low-altitude remote sensing small object detection tasks, FPNs serve as the core component for multiscale feature fusion. However, even though the backbone network has been lightened through DAMB, the cross-scale fusion process in FPNs remains a significant source of computational overhead. Taking RT-DETR's CCFF as an example, it performs cross-scale feature interaction via feature concatenation combined with convolutional fusion. Yet, this approach fails to fully

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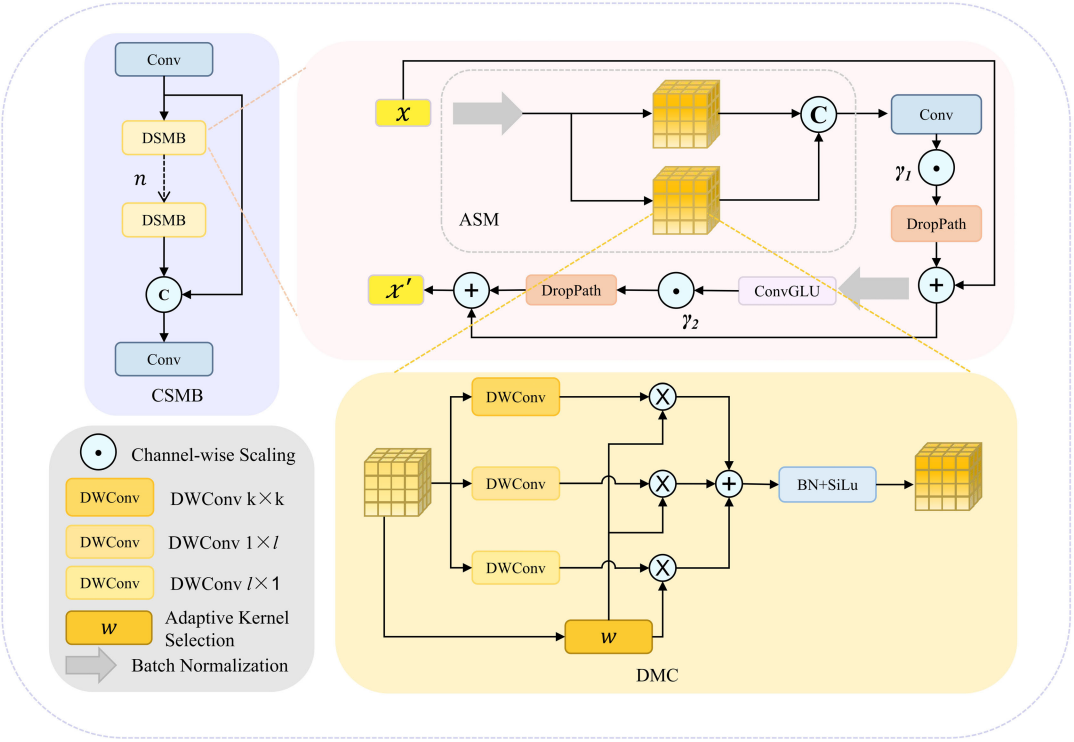


Fig. 2. Architecture of CSMB and its core components.

leverage high-level semantic information for selective filtering of low-level features during the fusion process, resulting in considerable computational redundancy. Meanwhile, conventional interpolation-based upsampling methods employ fixed sampling rules, lacking adaptability to feature content. Consequently, reducing network computational complexity while maintaining feature fusion effectiveness has become a critical challenge that demands immediate attention.

To address these issues, we propose LDFN, whose core principle lies in achieving network lightweighting while preserving multiscale fusion capabilities through hierarchical feature propagation pathways and content-adaptive sampling fusion mechanisms. The overall architecture is illustrated in Fig. 3. Distinguished from CCFF's feature concatenation strategy, LDFN replaces channelwise concatenation with modulation-based fusion: upsampled high-level features generate channelwise weights to adaptively filter semantically enhanced low-level features, followed by elementwise addition. This transforms cross-scale fusion from passive concatenation to active selection. Simultaneously, LDFN introduces dynamic point sampling [39] to replace fixed interpolation, achieving lightweight learnable upsampling through local offset prediction. For multiscale features output by the backbone, LDFN first applies channel semantic enhancement (CSE) to the highest level feature F_5, as shown in Fig. 3(a). The CSE module captures both global distribution and saliency response information of channels through dual-path pooling, then generates adaptive weights via a bottleneck structure to recalibrate features

$$F_5' = \sigma\left(f_c\left(AvgPool\left(F_5\right)\right) + f_c\left(MaxPool\left(F_5\right)\right)\right) \odot F_5 \quad (10)$$

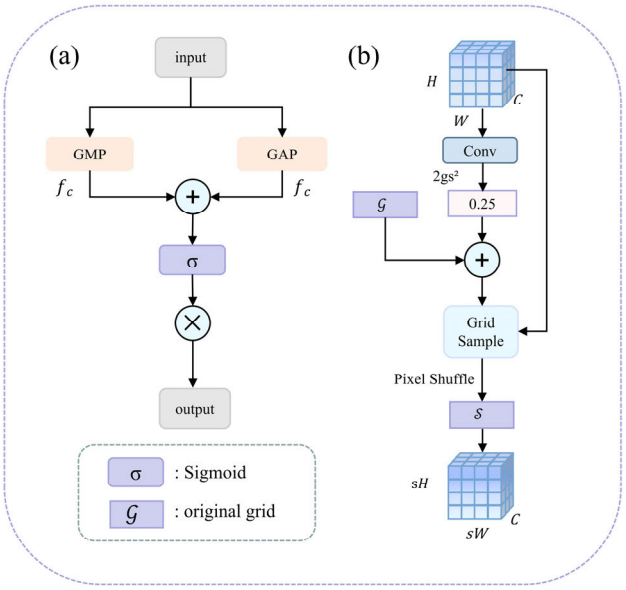


Fig. 3. Detailed structures of key components. (a) CSE module. (b) DS module.

where $AvgPool(\cdot)$ and $MaxPool(\cdot)$ denote global average pooling and global max pooling operations, respectively; $f_c(\cdot)$ represents the channel transformation function composed of two convolutional layers; and $\sigma(\cdot)$ denotes the sigmoid activation function.

In the feature resolution recovery stage, unlike nearest-neighbor or bilinear interpolation that adopt fixed sampling rules, LDFN formulates the upsampling process as a point

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sampling problem, determining optimal sampling positions by learning content-dependent offsets, as illustrated in Fig. 3(b). For input feature $X \in R^{C\times H\times W}$, offsets are generated through a lightweight $Conv_{1\times 1}$

$$\Delta P = \lambda \cdot Conv_{1\times 1}(X) + P_0 \tag{11}$$

where P_0 represents the initial sampling positions corresponding to bilinear interpolation, and λ denotes the offset range constraint factor. The upsampled features are obtained via differentiable grid sampling

$$Up(X) = GridSample(X, P_0 + \Delta P) \tag{12}$$

where GridSample$(\cdot,\cdot)$ represents the bilinear interpolation sampling function. This design ensures that the upsampling process requires only $O(Cgs^2)$ parameters, where g denotes the number of feature groups, significantly lower than the $O(C^2k^2)$ required by transposed convolution.

In the selective feature fusion stage, in contrast to CCFF's feature concatenation approach, LDFN utilizes upsampled high-level features to generate channel-adaptive weights for filtering and modulating channel-enhanced low-level features. The modulated features are then elementwise added to the upsampled features to obtain fused features. Taking the generation of P_4 as an example

$$P_4 = U_p\left(F_5'\right) + \sigma\left(f_c\left(U_p\left(F_5'\right)\right)\right) \odot \mathrm{Conv}_{1\times 1}(CSE(F_4)) \tag{13}$$

where $U_p(\cdot)$ denotes the dynamic point sampling upsampling operation, and $CSE(\cdot)$ represents the CSE operation. During the generation of P_3, LDFN employs the fused P_4 as the upsampling input, enabling shallow features to acquire richer multiscale information

$$P_3 = U_p(P_4) + \sigma\left(f_c\left(U_p(P_4)\right)\right) \odot \mathrm{Conv}_{1\times 1}(CSE(F_3)). \tag{14}$$

In summary, through the synergistic design of hierarchical feature propagation, adaptive feature filtering, and dynamic point sampling, LDFN constructs a lightweight multiscale feature fusion architecture. The selective fusion strategy enables high-level semantic information to effectively guide the filtering and retention of low-level features, while the dynamic sampling (DS) mechanism achieves content-adaptive feature upsampling with minimal parameter overhead. The combination of these components allows LDFN to effectively reduce network computational complexity while maintaining feature fusion capability.

D. ADGLs Function

Bounding box regression constitutes a critical component in object detection that directly determines localization accuracy, with the design of its loss function exerting substantial influence on both convergence speed and detection performance. In low-altitude remote sensing scenarios involving small object detection, where targets exhibit minimal dimensions and bounding box regression demonstrates high sensitivity to perturbations, more stringent requirements are imposed on the optimization capability of loss functions. Existing IoU-based loss functions primarily accelerate convergence through the incorporation of geometric penalty terms such as center point distance and shape constraints. However, these approaches overlook a fundamental limitation inherent to the IoU loss term itself: the inability to adaptively adjust optimization strength according to the quality of different regression samples.

To address this limitation, we propose ADGL by introducing an auxiliary bounding box mechanism coupled with efficient geometric constraints. This loss function builds upon a synergistic design framework integrating Inner-IoU [40] and EIoU [41], constructing auxiliary bounding boxes through scaling factors to compute IoU loss. This enables the network to generate differentiated gradient responses for regression samples of varying quality. Simultaneously, the shape loss is decomposed into two independent constraint terms for width and height, referred to as decoupled geometric constraints. This decoupling eliminates gradient conflicts during optimization and enables more precise constraints on geometric features.

The predicted and ground-truth boxes are denoted as B and B^{gt}, respectively, with center point coordinates (x_c, y_c) and (x_c^{gt}, y_c^{gt}), and dimensions w, h and w^{gt}, h^{gt}. ADGL first generates an auxiliary bounding box through the scaling factor *ratio*, with boundary coordinates defined as

$$\begin{aligned}
b_l^{gt} &= x_c^{gt} - \frac{w^{gt}\cdot ratio}{2}, \quad b_r^{gt} = x_c^{gt} + \frac{w^{gt}\cdot ratio}{2}\\
b_t^{gt} &= y_c^{gt} - \frac{h^{gt}\cdot ratio}{2}, \quad b_b^{gt} = y_c^{gt} + \frac{h^{gt}\cdot ratio}{2}\\
b_l^{aux} &= x_c - \frac{w\cdot ratio}{2}, \quad b_r^{aux} = x_c + \frac{w\cdot ratio}{2}\\
b_t^{aux} &= y_c - \frac{h\cdot ratio}{2}, \quad b_b^{aux} = y_c + \frac{h\cdot ratio}{2}.
\end{aligned} \tag{15}$$

The Inner-IoU is then computed based on the auxiliary bounding box

$$\begin{aligned}
inter = &\left(\min\left(b_r^{gt}, b_r^{aux}\right) - \max\left(b_l^{gt}, b_l^{aux}\right)\right)\cdot\left(\min\left(b_b^{gt}, b_b^{aux}\right)\right.\\
&\left.- \max\left(b_t^{gt}, b_t^{aux}\right)\right)
\end{aligned} \tag{16}$$

$$IoU^{inner} = \frac{inter}{(w^{gt}\cdot h^{gt})\cdot ratio^2 + (w\cdot h)\cdot ratio^2 - inter} \tag{17}$$

where $b_l^{aux}, b_r^{aux}, b_t^{aux}$ and b_b^{aux} denote the left, right, top, and bottom boundary coordinates of the auxiliary bounding boxes, respectively. Building upon this foundation, by integrating the Inner-IoU with the decoupled width–height shape constraints from EIoU, the proposed ADGL is formulated by incorporating center point distance loss and independent width–height shape penalty terms as follows:

$$\begin{aligned}
L_{Inner-EIoU} = 1 - IoU^{inner} &+ \frac{\rho^2\left(b, b^{gt}\right)}{(w^c)^2 + (h^c)^2}\\
&+ \frac{\rho^2\left(w, w^{gt}\right)}{(w^c)^2} + \frac{\rho^2\left(h, h^{gt}\right)}{(h^c)^2}
\end{aligned} \tag{18}$$

where $\rho(\cdot)$ denotes the Euclidean distance, b and b^{gt} denote the center points of the predicted box and ground-truth box, respectively; and w^c and h^c represent the width and height of the minimum enclosing rectangle covering both the predicted and ground-truth boxes. When $ratio < 1$, the auxiliary bounding box becomes smaller than the actual bounding

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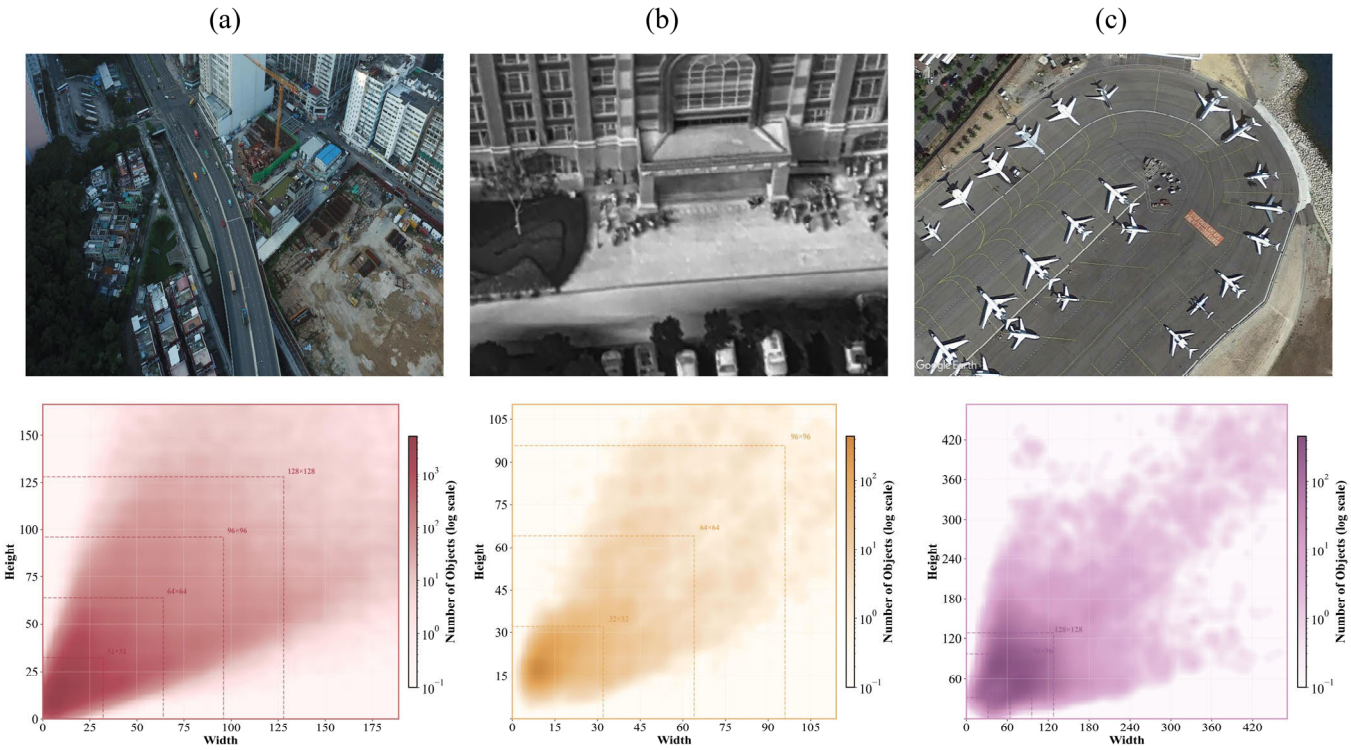


Fig. 4. Dataset visualization and target size distribution. (a) Sample images and size distribution from the VisDrone-DET2019 dataset. (b) Sample images and size distribution from the HIT-UAV dataset. (c) Sample images and size distribution from the SIMD dataset.

box, amplifying the gradient response for high-IoU samples and facilitating fine-grained localization. This focusing mechanism, operating synergistically with efficient geometric constraints, effectively enhances the localization accuracy for small objects in low-altitude remote sensing imagery.

IV. Experiments and Results Analysis

A. Datasets

To comprehensively evaluate the performance of the proposed method on low-altitude remote sensing small object detection tasks, four representative remote sensing datasets are selected for experimental validation: VisDrone-DET2019 [42], HIT-UAV [43], satellite imagery multivehicles dataset (SIMD) [44], and SODA-A [45], as illustrated in Fig. 4. These datasets encompass diverse imaging modalities and scene types—including visible light, infrared thermal imaging, and satellite remote sensing imagery—enabling thorough validation of the algorithm's detection accuracy, model efficiency, and generalization capability across diverse remote sensing scenarios.

VisDrone-DET2019 represents one of the largest UAV aerial object detection benchmarks currently available, comprising 10 209 visible light images captured from 14 cities across China. The dataset encompasses ten object categories with over 540 000 annotated instances, averaging approximately 53 objects per image.

HIT-UAV is the first publicly available high-altitude UAV infrared thermal imaging object detection dataset, containing 2898 infrared images that cover categories such as pedestrians, vehicles, and bicycles, totaling 24 899 annotated instances.

TABLE I
Experimental Details

Experimental environmental	Details
Operating system	Ubuntu 22.04
CPU	Intel Xeon Gold 6430
GPU	RTX 4090(24GB)
PyTorch version	PyTorch 2.1.0
CUDA version	CUDA 12.1
Programming language	Python 3.10
Batch size	4
Epochs	300
Optimizer	AdamW

SIMD is a satellite remote sensing dataset collected from Google Earth imagery across multiple countries. It comprises 5000 RGB images of resolution 1024×768, containing 45 096 annotated instances across 15 vehicle classes. The dataset covers diverse scene types and weather conditions, with dramatic object scale variations and high instance density per image, posing significant challenges for multiscale detection in remote sensing scenarios.

Small object detection dataset for aerial (SODA-A) scenarios is a large-scale aerial remote sensing benchmark dataset dedicated to small object detection in high-resolution imagery. Harvested from 2513 high-resolution aerial images, the dataset yields 16 653 images after preprocessing, with 872 069 instances exhaustively annotated across nine object categories, including small-vehicle, large-vehicle, ship, container, storage tank, swimming pool, airplane, windmill, and

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TABLE II
ABLATION EXPERIMENTS OF COMPONENTS ON VISDRONE DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Baseline	DAMB	LDFN	ADGL	Parameters	GFLOPs	mAP50	mAP50:95
✓				19.9	57	47.5	29.0
✓	✓			13.2	43.6	48.8	29.9
✓	✓	✓		10.9	37.6	49.1	30.1
✓	✓	✓	✓	**10.9**	**37.6**	**49.6**	**30.3**

TABLE III
ABLATION EXPERIMENTS OF COMPONENTS ON HIT-UAV DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Baseline	DAMB	LDFN	ADGL	Parameters	GFLOPs	mAP50	mAP50:95
✓				19.9	57	79.8	51.8
✓	✓			13.2	43.6	80.8	52.1
✓	✓	✓		10.9	37.6	80.8	52.0
✓	✓	✓	✓	**10.9**	**37.6**	**81.6**	**53.3**

helicopter. The dataset is characterized by extreme target density, significant scale variations, and complex backgrounds, representing the first large-scale remote sensing benchmark with exhaustive instance-level annotations tailored specifically for multicategory small object detection.

The rationale for selecting these four datasets lies in their shared characteristics typical of low-altitude remote sensing imagery, including challenges such as small object scales, large viewpoint variations, and complex backgrounds. Furthermore, the complementarity between visible light and infrared thermal imaging modalities enables validation of the method's cross-modal generalization capability.

B. Experimental Setup and Evaluation Metrics

All experiments in this study were conducted on a server equipped with an Intel Xeon Gold 6430 processor. An NVIDIA RTX 4090 GPU was utilized for model training and testing. The operating system was Ubuntu 22.04, with PyTorch 2.1.0 serving as the deep learning framework, Python 3.10 as the programming language, and CUDA version 12.1. Detailed configurations are provided in Table I.

To comprehensively evaluate the performance of the proposed method, evaluation metrics were selected across two dimensions: detection accuracy and model efficiency. In particular, mean average precision (mAP) was employed to measure detection accuracy, while parameters and GFLOPs were adopted to assess model efficiency. Higher mAP values indicate stronger detection performance, whereas lower parameters and GFLOPs signify a more lightweight network architecture and superior computational efficiency.

Precision (P) represents the proportion of true positives among all predicted positive samples, while recall (R) denotes the proportion of correctly predicted instances among all actual positive samples

$$P = \frac{TP}{TP + FP} \tag{19}$$

$$R = \frac{TP}{TP + FN} \tag{20}$$

where TP, FP, and FN represent the numbers of true positives, false positives, and false negatives, respectively. AP is obtained by integrating the area under the precision–recall curve, and mAP is calculated as the arithmetic mean of AP values across all categories

$$AP = \int_0^1 P(R)\, dR \tag{21}$$

$$mAP = \frac{1}{C} \sum_{i=1}^{C} AP_i \tag{22}$$

where C denotes the total number of object categories.

C. Ablation Experiments

To verify the effectiveness of each component in DAL-DETR, systematic ablation experiments were conducted on the VisDrone and HIT-UAV datasets. Detailed ablation results are presented in Tables II and III.

Upon introducing DAMB to the baseline model, dual optimization in both model compression and accuracy enhancement was achieved. The parameter count decreased from 19.9 M in the baseline to 13.2 M, representing a reduction of 33.7%, while computational cost dropped from 57 to 43.6 GFLOPs, decreasing by 23.5%. Moreover, detection accuracy also improved, with mAP50 on the VisDrone dataset increasing from 47.5% to 48.8% (a gain of 1.3% points) and mAP50:95 rising from 29.0% to 29.9%. On the HIT-UAV dataset, mAP50 increased from 79.8% to 80.8% (a gain of 1.0% point). These results demonstrate that DAMB achieves superior multiscale feature representation capability with fewer parameters through DMC and adaptive scale mixing mechanisms.

Subsequently, by replacing the traditional feature concatenation strategy with LDFN for cross-scale feature fusion, further compression was achieved on the already significantly reduced model. Although DAMB had reduced the parameter count from 19.9 to 13.2 M, LDFN successfully compressed it further to 10.9 M, eliminating an additional 17.4% of parameters.

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TABLE IV

COMPARISON OF DIFFERENT IOU LOSS FUNCTIONS ON VISDRONE DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	P	R	mAP50	mAP50:95
GIoU[46]	61.6	**48.2**	49.1	30.1
NWD-CIoU[47]	62.2	47.3	48.4	29.3
GCDIoU[48]	61.9	45.6	46.9	27.4
Inner-ShapeIoU[49]	61.8	47.7	48.7	29.8
Inner-EIoU	**63.0**	47.8	**49.6**	**30.3**

TABLE V

COMPARISON OF IOU LOSS FUNCTIONS ACROSS DIFFERENT DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Methods	mAP50	mAP50:95
VisDrone	GIoU[46]	49.1	30.1
	Inner-EIoU	**49.6**	**30.3**
HIT-UAV	GIoU[46]	80.8	52.0
	Inner-EIoU	**81.6**	**53.3**

Computational cost was further reduced from 43.6 to 37.6 GFLOPs, decreasing by an additional 13.8% (cumulatively, a 45.2% reduction in parameters and a 34.0% reduction in computational cost compared to the baseline). While substantially compressing the model size, detection accuracy was effectively preserved. On the VisDrone dataset, mAP50 reached 49.1%, and mAP50:95 reached 30.1%, representing improvements of 1.6% and 1.1% points over the baseline, respectively. On the HIT-UAV dataset, mAP50 reached 80.8%, and mAP50:95 reached 52.0%, similarly maintaining performance superior to the baseline. LDFN effectively preserves feature fusion quality by avoiding channel dimension expansion through modulation fusion, combined with a dynamic point sampling upsampling mechanism.

Finally, with the introduction of ADGL, the model achieved additional performance gains without increasing parameters or computational cost. On the VisDrone dataset, mAP50 reached 49.6% (an improvement of 2.1% points over the baseline), and mAP50:95 reached 30.3% (an increase of 1.3% points). On the HIT-UAV dataset, mAP50 reached 81.6% (an improvement of 1.8% points over the baseline), and mAP50:95 reached 53.3% (an increase of 1.5% points). Table IV presents comparative results against mainstream loss functions including GIoU [46], Nwd-CIoU [47], GCDIoU [48], and Inner-ShapeIoU [49], while Table V illustrates comparisons of loss functions across different datasets. These results demonstrate that ADGL achieves optimal performance on both datasets, validating its effectiveness in improving small object localization through auxiliary bounding box mechanisms and decoupled geometric constraints.

To further verify the rationale behind the kernel shape selection in DMC, we conduct ablation experiments on different kernel shape combinations, with results reported in Table VI. Here, A denotes the square kernel (k $\times$ k), B denotes the horizontal strip kernel (1 $\times$ l), and C denotes the vertical strip kernel (l $\times$ 1). As shown in the results, the model achieves the best mAP50 (48.8%) and mAP50:95 (29.9%) when all three geometrically complementary kernels are jointly employed, outperforming all other two-kernel

TABLE VI

ABLATION EXPERIMENTS OF DMC KERNEL CONFIGURATIONS ON VISDRONE DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

A	B	C	mAP50	mAP50:95
✓	✓		47.8	29.1
✓		✓	48.5	29.5
	✓	✓	48.3	29.2
✓	✓	✓	**48.8**	**29.9**

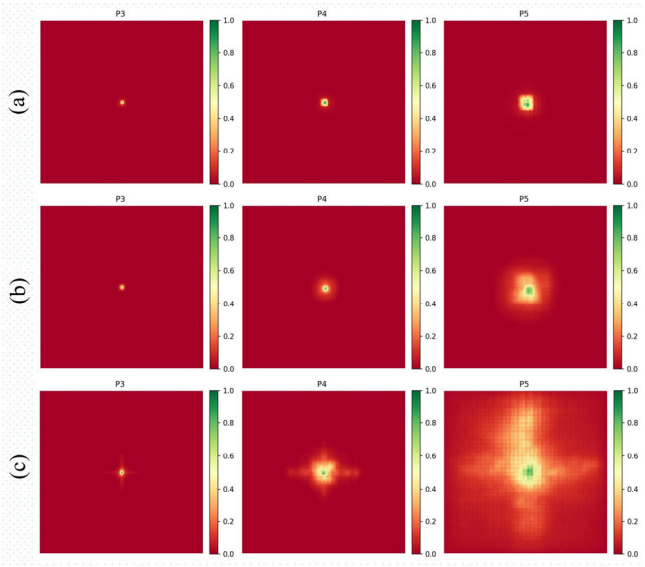


Fig. 5. ERF visualization comparison between standard convolution and DAMB. (a) YOLO26. (b) RT-DETR. (c) DAMB.

configurations. This indicates that the three kernel shapes extract features from complementary spatial orientations and collectively form a more complete multidirectional feature representation, confirming the rationality of the three-branch design adopted in DMC.

Beyond kernel shape selection, the feature extraction capability of DAMB from the perspective of receptive field coverage, we visualize and compare the effective receptive fields (ERFs) of different backbone networks, as illustrated in Fig. 5. The figure presents the ERF distributions of YOLO26 (a), the baseline RT-DETR (b), and the proposed DAMB (c) across three feature pyramid levels: P3, P4, and P5. It can be observed that YOLO26 (a) exhibits highly concentrated, dot-shaped ERF distributions at all levels, with rather limited spatial coverage. The baseline RT-DETR (b) shows moderately expanded ERFs compared to YOLO26, yet its overall distribution remains relatively compact. On the contrary, the proposed DAMB (c) presents notably different ERF characteristics: at the P3 level, the ERF extends in a distinct cross-shaped pattern, reflecting the ability of horizontal and vertical strip kernels to capture directional features; as the feature level deepens, the ERF coverage at P4 and P5 progressively broadens, with the P5 level achieving extensive coverage over the input space that far exceeds both YOLO26 and RT-DETR. These results confirm that DAMB, through multimorphology kernel fusion and the adaptive scale mixing mechanism, establishes a hierarchical receptive field system spanning from local to

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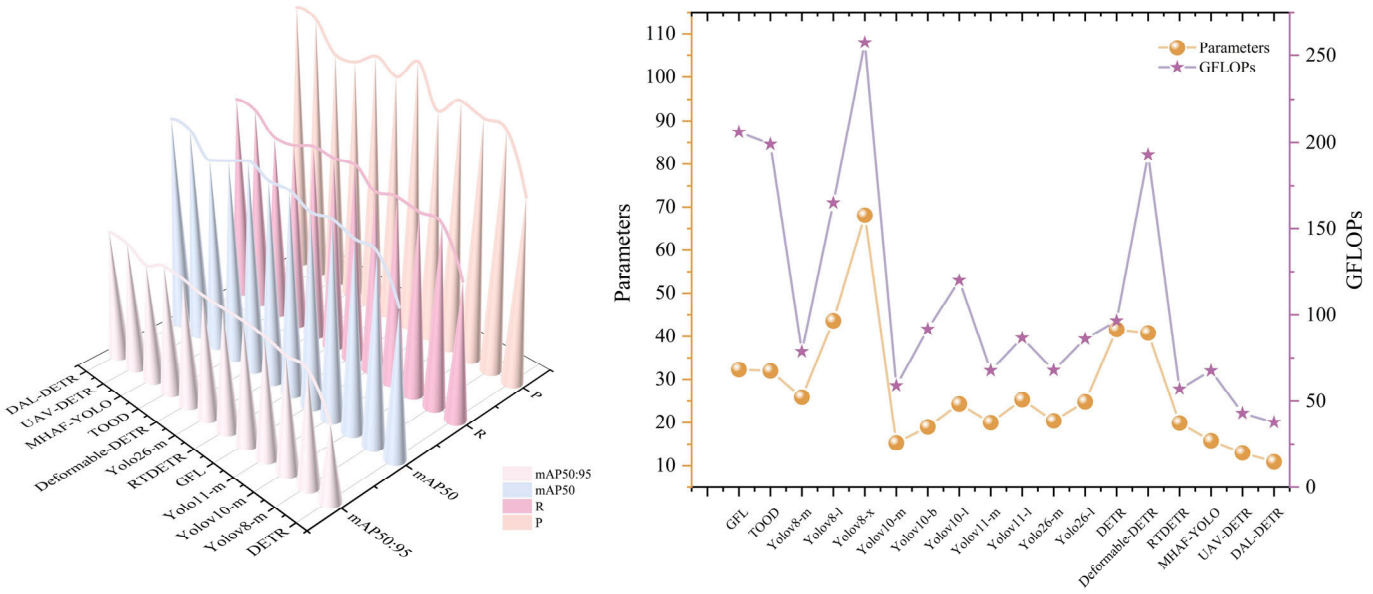


Fig. 6. Comparative visualization analysis of detection accuracy and model efficiency among state-of-the-art methods on the VisDrone-DET2019 dataset.

TABLE VII
COMPARISON WITH STATE-OF-THE-ART OBJECT DETECTION METHODS ON VISDRONE-DET2019 DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Model	Parameters	GFLOPs	P	R	mAP50	mAP50:95
GFL[50]	32.3	206	53.8	42.5	45.3	29.0
TOOD[51]	32.0	199	56.3	43.7	46.9	30.3
Yolov8-m[6]	25.9	78.7	56.3	42.9	44.4	27.1
Yolov8-l[6]	43.6	164.9	57.3	43.8	45.9	28.3
Yolov8-x[6]	68.1	257.4	57.3	45.3	46.7	29.0
Yolov10-m[7]	15.3	58.9	56.6	43.0	44.6	27.2
Yolov10-b[7]	19.0	91.7	57.9	43.3	45.7	28.1
Yolov10-l[7]	24.3	120	58.3	43.6	46.1	28.7
Yolov11-m[8]	20.0	67.7	58.2	43.8	46.2	28.5
Yolov11-l[8]	25.3	86.6	58.5	45.9	47.3	29.6
Yolo26-m[9]	20.4	67.9	57.1	45.4	47.3	29.1
Yolo26-l[9]	24.8	86.1	59.0	45.6	48.3	29.9
DETR[13]	41.6	96.5	43.9	32.7	35.1	18.3
Deformable-DETR[14]	40.8	193	59.6	45.7	49.2	30.0
MHAF-YOLO[26]	15.8	67.6	56.0	43.3	44.7	27.8
UAV-DETR[27]	13.0	42.9	62.2	47.4	49.3	30.2
RTDETR[16]	19.9	57	62.4	46.3	47.5	29.0
DAL-DETR	**10.9**	**37.6**	**63.0**	**47.8**	**49.6**	**30.3**

global scales, effectively strengthening the network's feature perception capability for targets of varying sizes.

D. Comparison Experiments

To comprehensively evaluate the performance of DAL-DETR, comparisons were conducted on the VisDrone dataset against various mainstream object detection methods, including anchor-based single-stage detectors (GFL [50], TOOD [51]), the YOLO series (YOLOv8 [6], YOLOv10 [7], YOLOv11 [8], YOLO26 [9]), and Transformer-based end-to-end detectors (DETR [13], Deformable-DETR [14], RT-DETR [16]). All competing methods were implemented with their official codebases and retrained on the VisDrone-DET2019 dataset under the same data split and hardware platform. Each method was trained according to the training configuration specified in its official implementation. Experimental results are presented in Table VII and Fig. 6.

Although anchor-based single-stage detectors (GFL, TOOD) perform well in general object detection tasks, they face evident efficiency challenges in low-altitude remote sensing scenarios. These methods typically have parameter counts exceeding 32 M and computational costs approaching 200 GFLOPs, yet their mAP50 values remain below 47%. Even though TOOD achieves an mAP50:95 of 30.3%, matching that of DAL-DETR, its model size remains substantially larger than the proposed method, indicating that traditional architectures

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require considerable model capacity to attain acceptable performance when handling small objects.

As representative real-time detection methods, the YOLO series exhibits tradeoffs between accuracy and efficiency across different scale configurations. Lightweight configurations (YOLOv8-m, YOLOv10-m, and YOLOv11-m) have parameter counts ranging from 15.3 to 25.9 M, yet their mAP50 values all fall below 47%. While medium and larger configurations can improve accuracy, they substantially increase both parameter count and computational cost. For instance, YOLOv8-x with 68.1-M parameters and 257.4 GFLOPs achieves only 46.7% mAP50.

Among methods specifically optimized for small object detection, MHAF-YOLO and UAV-DETR are also representative. Although MHAF-YOLO is designed for small object scenarios, its mAP50 of only 44.7% is significantly lower than that of DAL-DETR, indicating that its feature extraction strategy still lacks sufficient capability to capture small objects under complex backgrounds. UAV-DETR, as a Transformer-based detector specifically optimized for small object detection in low-altitude remote sensing, achieves detection accuracy closest to DAL-DETR among all compared methods, yet still falls behind DAL-DETR across all metrics including mAP50, mAP50:95, and parameter count, further demonstrating the superior balance between detection accuracy and model lightweighting achieved by the proposed method in low-altitude remote sensing small object detection scenarios. Notably, as the most recently proposed method specifically optimized for small objects, YOLO26 achieves mAP50 values of 47.3% and 48.3% in its m and l configurations, respectively, representing the best performance within the YOLO series. However, compared to DAL-DETR, YOLO26-l still has over twice the parameters and computational cost while delivering 1.3% points lower mAP50, highlighting the advantages of the proposed method in lightweight design.

Transformer-based end-to-end detectors exhibit distinct performance characteristics. The original DETR achieves only 35.1% mAP50 due to slow training convergence. Deformable-DETR improves mAP50 to 49.2% through deformable attention, but at the cost of 40.8-M parameters and 193 GFLOPs computational overhead. The baseline RT-DETR achieves better efficiency while maintaining end-to-end advantages, reaching 47.5% mAP50 with 19.9-M parameters and 57 GFLOPs.

Compared to the aforementioned methods, DAL-DETR achieves optimal balance between parameter efficiency and detection accuracy. Relative to the baseline RT-DETR, the proposed method reduces parameters by 45.2% and computational cost by 34.0% while improving mAP50 to 49.6%, representing an increase of 2.1% points.

The comparative results demonstrate that DAL-DETR achieves an optimal tradeoff between parameter efficiency and detection accuracy. In particular, DAMB substantially enhances multiscale feature representation through its multimorphology receptive field fusion mechanism; building upon this foundation, LDFN further compresses the model by employing modulation-based fusion, effectively reducing the computational cost of cross-scale integration. Meanwhile, ADGL improves localization accuracy for small objects via its auxiliary bounding box mechanism. The synergy of these components enables DAL-DETR to achieve state-of-the-art performance in low-altitude remote sensing small object detection.

E. Visualization Experiments

To intuitively demonstrate the performance advantages of DAL-DETR in low-altitude remote sensing small object detection tasks, we conduct visualization analysis from two perspectives: detection outputs and feature representations. To compare detection capabilities across different methods in complex low-altitude remote sensing scenarios, we present comparative detection results of DAL-DETR against the baseline models RT-DETR and YOLO26 on the VisDrone-DET2019 dataset (see Fig. 7), illustrating improvements in typical issues such as missed detections and false positives. Furthermore, we extend the qualitative evaluation to the SODA-A dataset (see Fig. 8), where comparisons against RT-DETR reveal DAL-DETR's ability to suppress misclassification errors and redundant predictions in complex aerial scenes. To further analyze differences in feature extraction across methods, we employ the GradCAM technique to generate feature heatmaps, comparing feature response patterns on both the VisDrone visible-light dataset (see Fig. 9) and HIT-UAV infrared dataset (see Fig. 10), thereby validating DAL-DETR's feature focusing capability and cross-modal generalization performance. Each heatmap is independently normalized on a per-image basis to better visualize the spatial distribution of feature activations across different images.

Fig. 7 presents comparative detection results across various typical low-altitude remote sensing scenarios on the VisDrone dataset. The primary limitation of the baseline RT-DETR lies in its elevated false positive rate, generating numerous erroneous detections in background regions including road markings, shadow edges, and building textures. While YOLO26 is specifically optimized for small object detection, it suffers from substantial missed detections, particularly in dense target regions and occlusion scenarios where numerous objects remain undetected. On the contrary, DAL-DETR demonstrates superior detection performance across all scenarios: the selective fusion mechanism in LDFN effectively suppresses background noise interference, significantly reducing false positives, while the multimorphology receptive field fusion mechanism in DAMB enhances the capture capability for densely distributed and occluded small objects, markedly decreasing missed detections. This advantage becomes particularly pronounced under challenging conditions such as complex illumination.

To further validate the generalization of DAL-DETR's detection reliability beyond the VisDrone benchmark, we present additional qualitative results on the SODA-A dataset in Fig. 8. As shown, RT-DETR exhibits notable limitations, including misclassification errors and redundant overlapping predictions, particularly in dense scenes. For instance, in the residential parking scenario, RT-DETR incorrectly classifies a target as "container," while in the airport and suburban scenes, excessive overlapping bounding boxes indicate a high false

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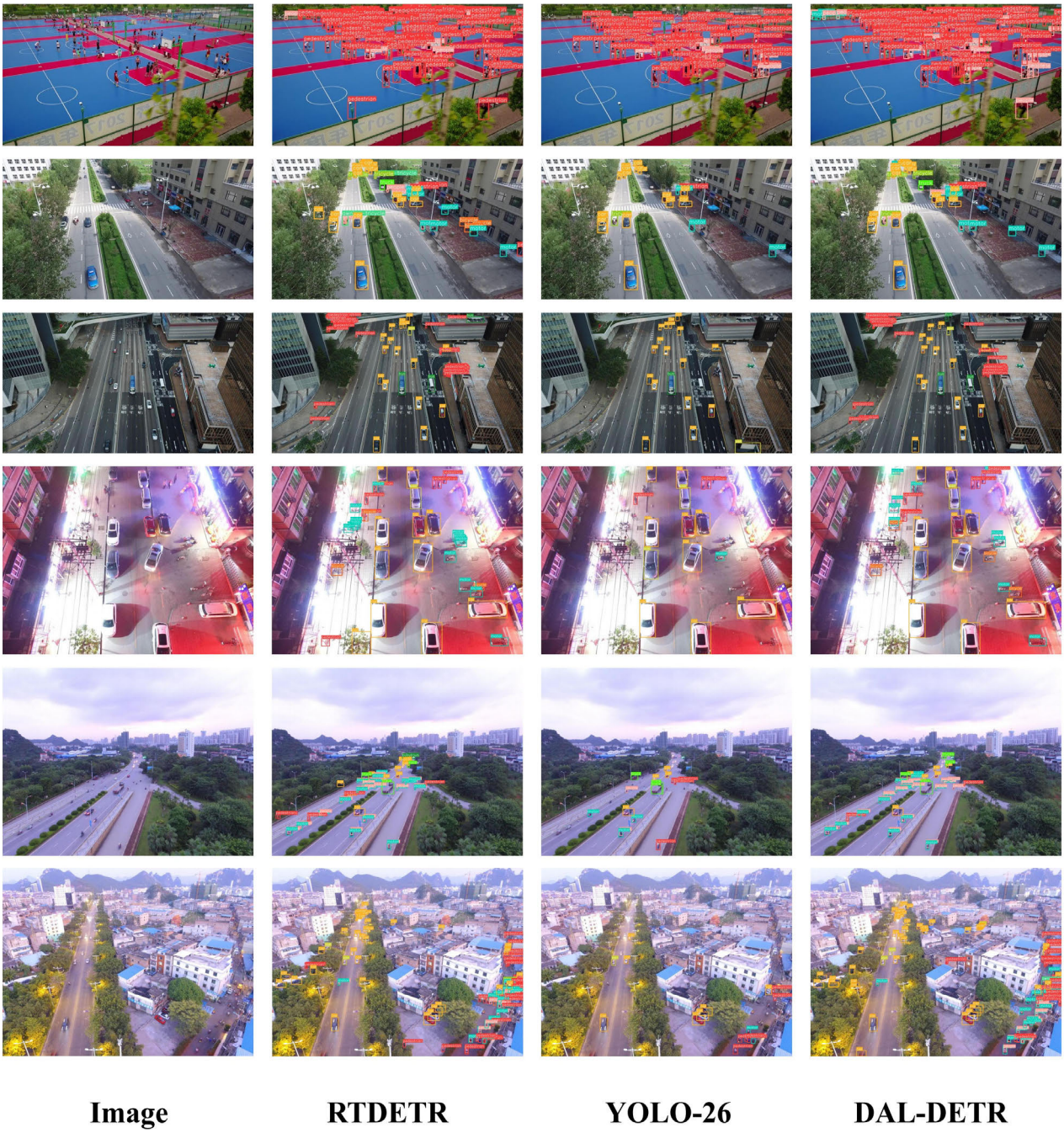


Fig. 7. Qualitative detection results on the VisDrone-DET2019 dataset. Comparison of detection performance among RT-DETR (baseline), YOLO26, and DAL-DETR across various challenging scenarios.

positive rate. On the contrary, DAL-DETR produces cleaner and more precise detections across all scenarios, effectively suppressing misclassifications and redundant predictions. This further demonstrates that the selective fusion mechanism in LDFN and the discriminative feature representations learned via DAMB jointly contribute to improved detection reliability across diverse complex aerial scenes.

Heatmap comparisons on the VisDrone dataset reveal that the baseline model exhibits relatively dispersed responses, with insufficient discrimination between target and background regions. Notably, certain background areas display strong activation responses, corresponding to the observed false positive patterns. Conversely, DAL-DETR demonstrates enhanced focusing characteristics, with high-response regions concentrated at target locations while background activation is effectively suppressed. This indicates that the dynamic weight generation mechanism in DAMB enables the network to learn more discriminative feature representations, as illustrated in Fig. 9.

Heatmap analysis on the HIT-UAV infrared dataset further validates the cross-modal generalization capability of our approach. Infrared imagery presents inherent challenges,

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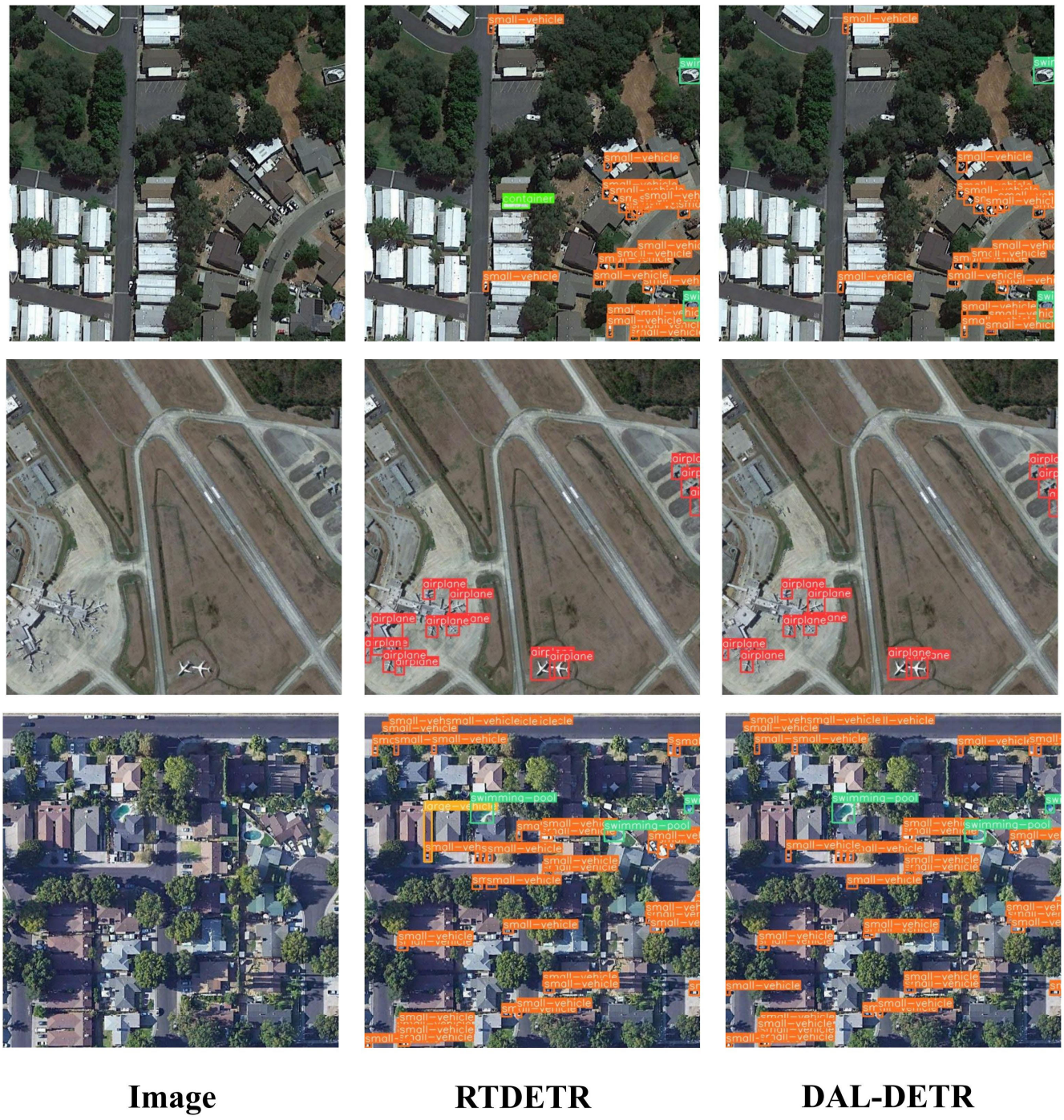


Fig. 8. Qualitative detection results on the SODA-A dataset. Comparison of detection performance between RT-DETR (baseline) and DAL-DETR across various challenging scenarios.

including low target-background contrast and blurred edges. The baseline model exhibits even more dispersed responses in infrared scenarios, particularly struggling to distinguish targets from background in low-contrast scenes; substantial room for improvement remains. Heatmap comparisons reveal that, while the baseline model exhibits somewhat dispersed responses in infrared scenarios, DAL-DETR accurately localizes target positions, forming concentrated high-response regions with well-defined boundaries. These results demonstrate that through hierarchical multiscale representation and selective feature fusion mechanisms, DAL-DETR not only achieves further performance enhancement in infrared scenarios, but its feature enhancement strategy also exhibits strong modality-agnostic properties, maintaining stable performance across different imaging conditions, as shown in the comparative heatmaps in Fig. 10.

In summary, the visualization experiments validate the effectiveness of DAL-DETR from both detection outputs and

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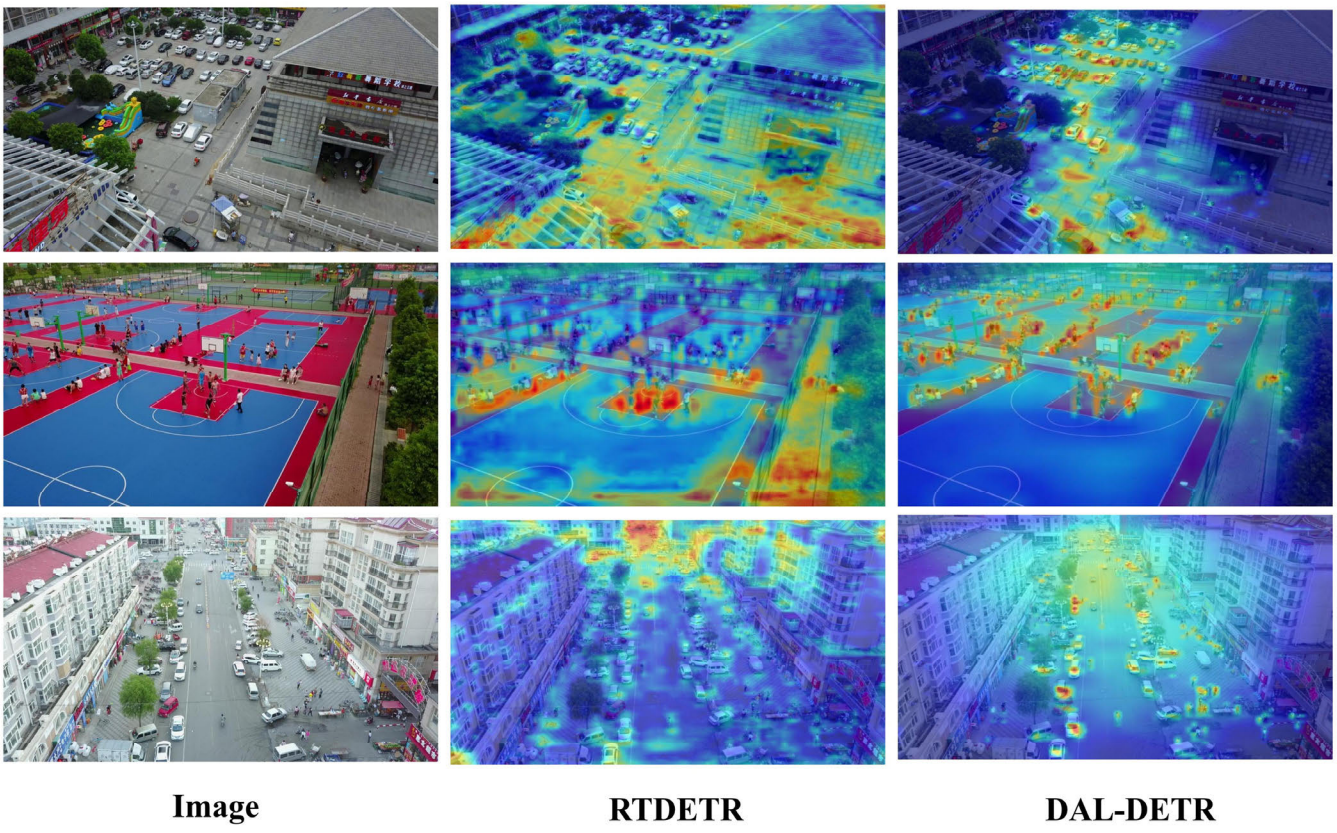


Fig. 9. Feature attention visualization on the VisDrone-DET2019 dataset. Comparison of feature activation heatmaps between RT-DETR and DAL-DETR.

TABLE VIII
GENERALIZATION EVALUATION OF THE BASELINE AND DAL-DETR ON MULTIPLE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Methods	Parameters	GFLOPs	mAP50	mAP50:95
VisDrone[42]	RTDETR	19.9	57.0	47.5	29.0
	DAL-DETR	**10.9**	**37.6**	**49.6**	**30.3**
HIT-UAV[43]	RTDETR	19.9	57.0	79.8	51.8
	DAL-DETR	**10.9**	**37.6**	**81.6**	**53.3**
SIMD[44]	RTDETR	19.9	57.0	77.4	62.0
	DAL-DETR	**10.9**	**37.6**	**79.9**	**64.6**
SODA-A[45]	RTDETR	19.9	57.0	72.6	42.2
	DAL-DETR	**10.9**	**37.6**	**73.5**	**42.8**

feature representations. Across various challenging scenarios, DAL-DETR consistently demonstrates superior detection capability and more precise feature focusing: compared to baseline methods, our approach significantly reduces both false positives and false negatives, with feature activations concentrated on target regions while background interference is effectively suppressed. These results indicate that DAL-DETR learns more discriminative feature representations, confirming the efficacy of our lightweight design strategy.

F. Generalization Performance

To evaluate the cross-dataset generalization capability of DAL-DETR, experiments were conducted on four datasets: VisDrone, HIT-UAV, SIMD, and SODA-A. These datasets cover diverse imaging modalities and scene conditions, offering a broad basis for evaluating model adaptability in low-altitude remote sensing. Results are presented in Table VIII.

As illustrated in Table VIII, DAL-DETR yields consistent gains over the RT-DETR baseline across all four datasets. On VisDrone, mAP50 increases by 2.1% points to 49.6%. On HIT-UAV, mAP50 and mAP50:95 reach 81.6% and 53.3%, with gains of 1.8% and 1.5% points over the baseline, respectively. On SIMD and SODA-A, DAL-DETR likewise outperforms the baseline in both metrics. These gains are obtained with a 45.2% reduction in parameters and a 34.0% decrease in computational cost relative to the baseline.

The consistent improvements across four datasets with varying imaging conditions and scene distributions suggest that the proposed method generalizes reliably within the evaluated low-altitude remote sensing settings.

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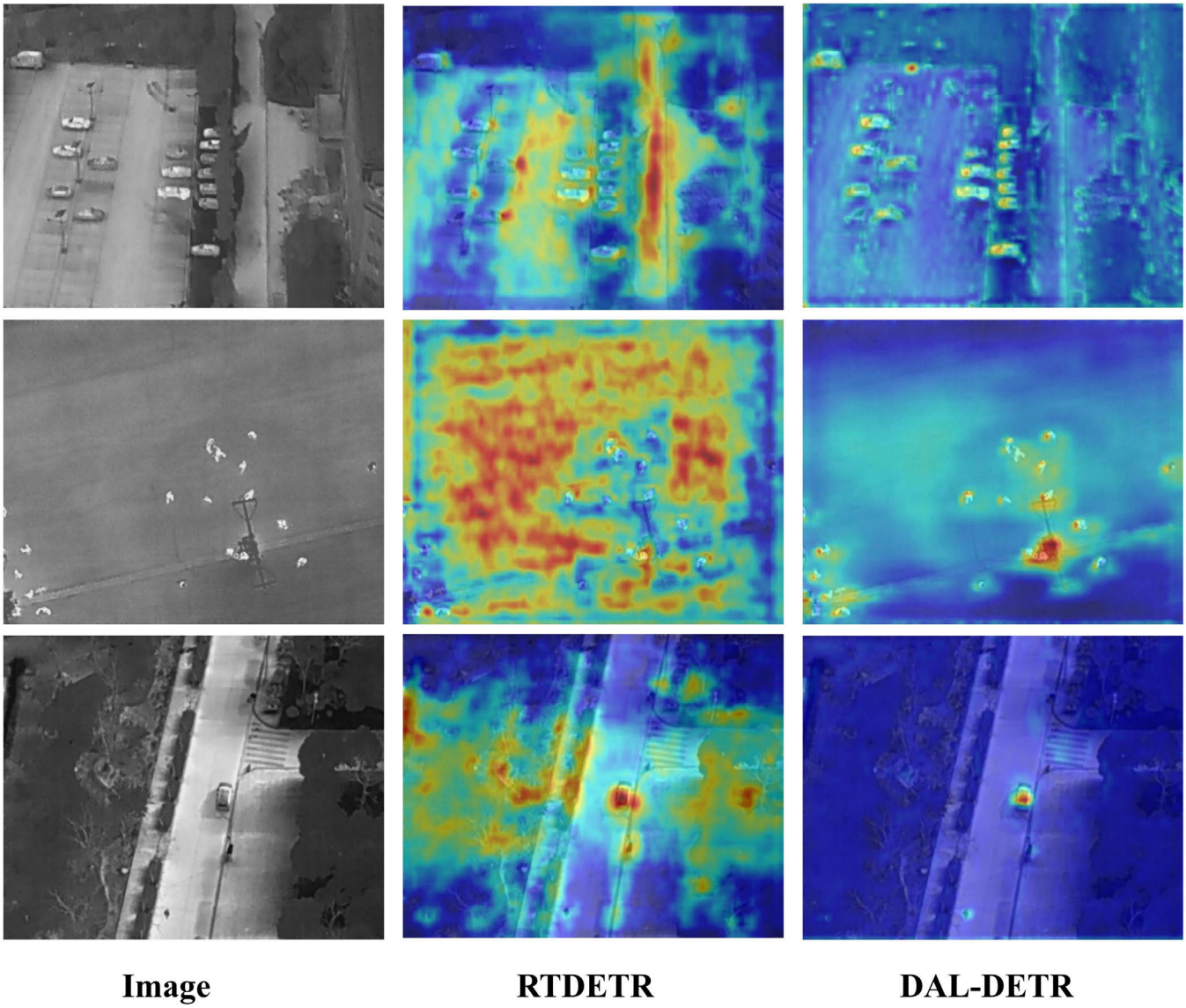


Fig. 10. Feature attention visualization on the HIT-UAV dataset. Comparison of feature activation heatmaps between RT-DETR and DAL-DETR in infrared imaging scenarios.

TABLE IX
INFERENCE EFFICIENCY COMPARISON BETWEEN BASELINE AND DAL-DETR

Models	Device	Model size (MB)	Memory (MB)	Latency （ms）
RTDETR	GPU	77.0	178.1	18.1
DAL-DETR	GPU	42.4	156.0	19.0
RTDETR	CPU	77.0	/	130.7
DAL-DETR	CPU	42.4	/	121.1

G. Inference Efficiency and Accuracy Tradeoff Analysis

To comprehensively assess the practical computational efficiency of DAL-DETR, inference latency, peak GPU memory consumption, and model storage size were measured under both GPU and CPU execution environments. The results are summarized in Table IX.

On GPU, DAL-DETR compresses model storage from 77.0 to 42.4 MB, a reduction of 44.9%, and lowers peak VRAM usage from 178.1 to 156.0 MB, with only a 0.9 ms increase in inference latency relative to the baseline. This overhead stems from the irregular memory access patterns of the dynamic point sampling and deformable modulation operators, which are inherently less suited to standard GPU parallelization. That said, for hardware platforms operating under tight computational budgets, memory footprint and storage capacity tend to be more binding constraints than raw GPU throughput—the meaningful reductions achieved by DAL-DETR on both fronts therefore carry greater practical weight.

On CPU, inference latency drops from 130.7 to 121.1 ms, a 7.3% improvement. This gain more directly reflects the benefits of the lightweight architecture than the GPU results do: without the memory access bottlenecks introduced by dynamic operators, the reductions in parameter count and computational cost translate more fully into measurable speed

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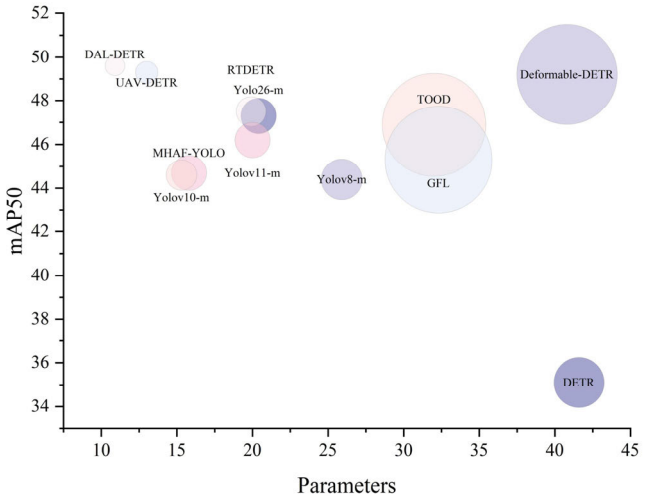


Fig. 11. Accuracy-efficiency tradeoff comparison on the VisDrone-DET2019 dataset. Bubble size indicates GFLOPs.

gains. It also better approximates conditions found in hardware environments without dedicated accelerators. Taken together, the measurements across both execution environments indicate that DAL-DETR strikes a sound balance between model compactness and inference efficiency.

Fig. 11 further presents a bubble chart on the VisDrone-DET2019 dataset to illustrate the tradeoff between efficiency and accuracy, where bubble size reflects GFLOPs. DAL-DETR occupies the upper left region with the smallest bubble, indicating that it simultaneously achieves the fewest parameters, lowest computational cost, and highest detection accuracy among all compared methods. Among more complex architectures, Deformable-DETR sits in the upper right region, where its accuracy falls slightly below that of DAL-DETR while demanding greater parameters and GFLOPs; DETR likewise resides on the right side of the figure yet yields considerably lower accuracy, placing it in the lower right region. These results demonstrate that DAL-DETR strikes the best balance between detection accuracy and model efficiency across all evaluated methods.

H. Limitations of the Proposed Method

We further discuss the limitations of the proposed method. Regarding complex imaging conditions, the orthogonal strip kernel design in DAMB enables effective capture of horizontal and vertical structural features (see Fig. 5); however, directional coverage remains limited for targets at large oblique angles and under severe perspective distortion, and systematic validation under such conditions is left for future work. Regarding oriented object detection, arbitrary-orientation targets are prevalent in low-altitude remote sensing scenes, and oriented bounding boxes offer more precise geometric descriptions of such objects than their axis-aligned counterparts. That said, most existing low-altitude remote sensing datasets rely on horizontal bounding box annotations, leaving oriented detection benchmarks relatively scarce; oriented detection methods also typically require dedicated angle prediction branches and oriented box regression mechanisms, which introduce considerable model complexity. Extending the proposed framework to oriented object detection while preserving model efficiency therefore remains an important direction for future investigation.

V. Conclusion

Addressing the challenge of balancing model compactness and detection accuracy in low-altitude remote sensing small object detection, this study proposes DAL-DETR, an efficient network tailored for low-altitude remote sensing object detection. The network achieves efficient integration of multiscale receptive fields through DAMB, employs modulated fusion via LDFN in place of feature concatenation, and incorporates ADGL to enhance small object localization accuracy. These designs simultaneously reduce model complexity and improve detection performance, achieving an effective balance between lightweight architecture and accuracy.

Experiments on the VisDrone, HIT-UAV, SIMD, and SODA-A datasets demonstrate that, with a 45.2% reduction in model parameters and a 34.0% decrease in computational cost, DAL-DETR still achieves substantial improvements in detection accuracy, outperforming the baseline model and various state-of-the-art methods. These results validate the effectiveness and generalization capability of the proposed approach.

Future work will advance in three directions. First, we will extend the proposed dynamic adaptive mechanism and selective fusion strategy to downstream tasks such as object tracking and instance segmentation, establishing a unified lightweight visual perception framework. Second, we will enhance the model's adaptability to challenging imaging conditions, including multimodal remote sensing scenarios and arbitrarily oriented object detection, to further broaden its applicability in complex low-altitude remote sensing environments. Third, fine-grained analysis of inference bottlenecks and the exploration of faster operator approximations for GPU-speed-critical deployment scenarios will be investigated.

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